

GPUS AND COOLING SYSTEM THEORY

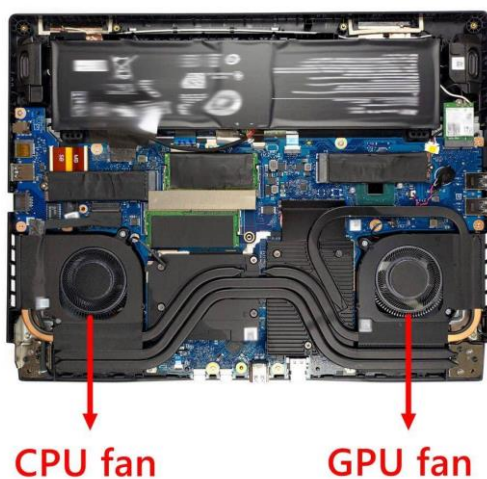
- 1) **Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.**

❖ GPU Details:

Component	Details
GPU Brand	NVIDIA
GPU Model	NVIDIA GeForce RTX 4050 Laptop GPU
Type	Dedicated GPU
Integrated GPU	Intel Iris Xe Graphics (Integrated)

- 2) **Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.**

❖ FAN



❖ HEATSINK



Cooling Components

- I. **Cooling Fan**
 - Removes hot air from inside the system.
 - Helps maintain safe operating temperature.
 - II. **Heatsink**
 - A metal component that absorbs heat from the GPU/CPU.
 - Transfers heat toward the fan for cooling.
 - III. **Heat Pipe**
 - Transfers heat from GPU/CPU to the heatsink area.
 - IV. **Thermal Paste**
 - Improves heat transfer between GPU/CPU and heatsink.
- 3) **Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.**

❖ Air Cooling vs Liquid Cooling for GPUs

Factor	Air Cooling	Liquid Cooling
Cost	Cheaper and budget-friendly	More expensive
Cooling Efficiency	Good for normal gaming and daily use	Better for high-performance gaming and overclocking
Maintenance	Low maintenance, only cleaning dust	Requires checking liquid pump and coolant
Noise Level	Fans can become louder under heavy load	Usually quieter with proper setup
Installation	Easy to install	More complex installation

4) Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research

- I would choose liquid cooling for my GPU because it provides excellent heat dissipation during high-performance gaming and streaming. It keeps the GPU temperature lower, improves system stability, and helps maintain smooth performance during long IPL streaming and gaming sessions. Although it is more expensive than air cooling, its efficiency makes it suitable for a powerful gaming PC.